

BhashaGuard Edge

On-Device Cultural Hallucination Guard for Indian-Language AI

An offline review layer that checks AI responses against local evidence and explains what remains uncertain.

Snapdragon® AI Lab Build & Present Challenge 2026

Solution proposal. No measured accuracy or device-performance results claimed.

Problem and proposed contribution

The review problem

An answer can use the correct script while confusing a local practice, place or cultural context.

The proposed guard

Highlight questionable claims and show the relevant source passages, with explicit abstention.

First user

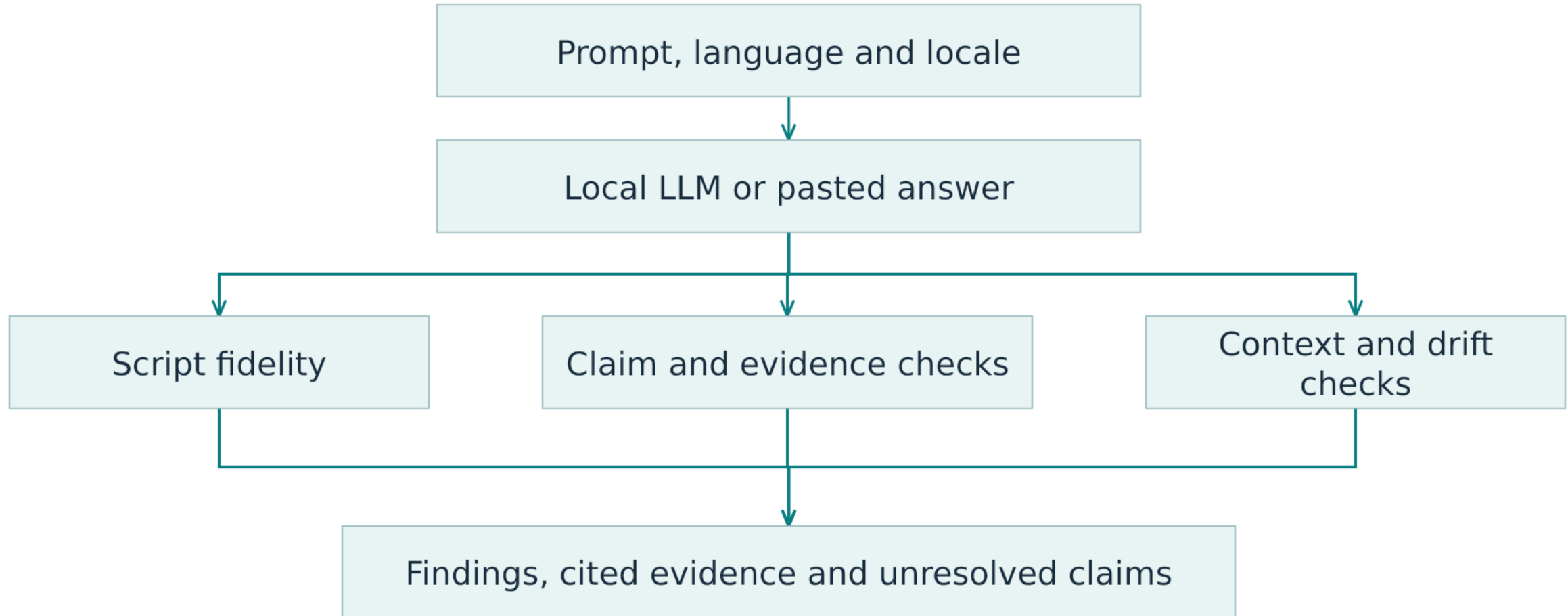
A teacher reviewing an AI-written Telugu learning note before sharing it with students.

What distinguishes the design

Locale-qualified evidence and valid regional variants, supported by Dravidian-language evaluation.

Initial scope: Telugu text. Tamil and Kannada follow native-speaker validation.

Local checking architecture



Reviewed local evidence pack: sources, locale, date, licences and valid variants.

Snapdragon deployment plan

Target: Snapdragon-powered HP PC running Windows ARM64

Stage	Candidate	Execution plan
Generation / verification	Qwen3-4B-Instruct-2507	Matching QAIRT / GenieX bundle, after exact-device validation
Retrieval / drift	multilingual-e5-small	CPU baseline, then a proposed custom ONNX / QNN port
Script / evidence / UI	Unicode, SQLite, local service	CPU processing and local storage

Validate language quality, quantization and HTP execution traces.
Keep private drafts off remote profiling jobs and test offline operation.

Model recipes exist. Exact HP-PC compatibility and NPU gains remain to be measured.

Evaluation plan

Checking quality

Native-speaker labels with adjudication. Hold out topics and source families to reduce leakage.

Reported outcomes

Macro-F1, valid-variant false alarms, retrieval recall, citation validity and abstention versus error.

Script fidelity = $\text{target-script letters} / \text{assessed letters}$

Evidence consistency = $\text{supported} / (\text{supported} + \text{contradicted})$

Report evidence coverage alongside consistency. Zero denominators return N/A.

Device performance

Measure median / p95 guard latency, peak memory, model size and equivalent CPU / NPU runs.

Controls

Compare script-only, retrieval-only and ungrounded judging with the full guard.

Demonstration and delivery

Planned teacher workflow

1. Select Telugu and a locale, then analyse a labelled test paragraph.
2. Inspect a script error, a contradiction and an unresolved claim.
3. Open cited evidence and check a valid regional-variant example.
4. Export the report with actual backend details and runtime timings.

Build sequence

Telugu CPU prototype, followed by validated Snapdragon inference.
Extend to Tamil and Kannada after language-specific evaluation.

Limits: incomplete evidence and verifier errors require abstention and human review.

Technical references

1. **Qualcomm Qwen3-4B recipe**
2. GenieX runtime documentation
3. Multilingual E5 small model card
4. ONNX Runtime QNN execution provider
5. AI Hub compilation documentation
6. Unicode script properties
7. BHRAM-IL prior research

Source URLs are included in speaker notes and in the companion project description.